AATIF NISAR DAR

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RESEARCH INTERESTS

Information and Communication Technologies for Development (ICTD), Internet Measurement, Remote Sensing, and AI/ML.

EDUCATION

PhD in Computer Science *2025 - Present*
Virginia Tech, Blacksburg VA, USA GPA: 4.0/4.0

Remote Sensing Interdisciplinary Graduate Education Program *2025 - Present*
Virginia Tech, Blacksburg VA, USA

Master of Science in Computer Science *2020 - 2022*
South Asian University, New Delhi, India CGPA: 7.87/9.0 (Second Rank)

Bachelor of Science (Honours) in Computer Science *2017 - 2020*
University of Delhi, New Delhi, India CGPA: 7.70/10.0

FELLOWSHIPS/AWARDS AND HONORS

- Remote Sensing Interdisciplinary Graduate Education Program Fellowship, First-Year PhD Fellowship *Aug'25 - Jun'26*
- Best Presenter Award, IEEE ICETCI 2023 *2023*

PUBLICATIONS

Research Papers

Aatif Nisar Dar et al. "Broadband Surveyor Tool (BBS): Surveying Long-Tail ISPs." *Research with Dr. Shaddi Hasan; under submission to ACM Internet Measurement Conference (IMC '27).*

Aatif Nisar Dar et al. "If You Build It, They Will Come: Auditing Broadband Deployment via Crowdsourced Network Measurements." *Research with Dr. Shaddi Hasan; under submission to ACM Internet Measurement Conference (IMC '27).*

Aatif Nisar Dar et al. "A Methodology to Assess Equity in Access to Water in Villages in India." *Research with Prof. Aaditeshwar Seth; manuscript in preparation.*

Raman Kumar, **Aatif Nisar Dar**, Aaditeshwar Seth. "Multi-Resolution Satellite Data Processing for Contextually Relevant Land Use-Land Cover Mapping: Distinguishing Scrublands, Farms, and Plantations Across India." *Under submission to the AAAI-27 Artificial Intelligence for Social Impact (AISI) Special Track.*

Aatif Nisar Dar, Aditya Raj Singh, Anirban Sen. "PolicyStory: Leveraging Large Language Models to Generate Comprehensible Summaries of Policy-News in India." Accepted at the Proceedings of the 2025 International Conference of The Computational Social Science Society of the Americas

Aatif Nisar Dar, Nandana Sengupta, Chetan Arora. "Assessing the Feasibility and Ethics of Economic Status Prediction using Deep Learning on Household Images." ACM Journal on Computing and Sustainable Societies (2024).

Aatif Nisar Dar and Reshma Rastogi, 2023. "MLGAN: Addressing Imbalance in Multilabel Learning Using Generative Adversarial Networks," In Proceedings of IEEE International Conference on Emerging Techniques in Computational Intelligence (ICETCI), Hyderabad, India, 2023, pp. 324-331. ***Best Presenter Award***

Aatif Nisar Dar. "Principal Component Analysis," Global Scientific Journal, GSJ: Volume 9, Issue 7, July 2021, Online: ISSN 2320-9186. (*Lightly Peer-reviewed*)

Posters/Workshops/Symposiums

Aatif Nisar Dar, Jocelyn Bliton, Shaddi Hasan. "If You Build It, They Will Come: Auditing Broadband Deployment via Crowdsourced Network Measurements." Poster presented at the Telecommunications Policy Research Conference (TPRC '26), Washington, D.C., 2026.

Aatif Nisar Dar, Shaddi Hasan. "Capturing Internet Service Provider (ISP) Footprints: Multi-Modal Spatio-Temporal Representations for Broadband Infrastructure." Poster at OGIS (Office of Geographic Information Systems and Remote Sensing Research), Virginia Tech, 2026.

Aatif Nisar Dar, Badrinath Padmanabhan, Mrunal Atul Kadane, Aaditeshwar Seth. "Using Satellite Images to Track Relative Socioeconomic Development in Rural India". Poster at ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS '24), 2024.

TPRC Graduate Student Workshop, Washington, D.C. ***Funded travel*** *24 Sep '26*
Research Week with Google 2024. ***Funded travel*** *1–3 Feb '24*
Symposium on Machine Learning, IIT Bombay. ***Funded travel*** *21–23 Dec '23*

EXPERIENCE

Graduate Research Assistant, Virginia Tech. *Aug '25 - Present*
SPIN Lab
PI: Dr. Shaddi Hasan

1. Auditing broadband deployment using crowdsourced network measurements.

- Developed methods to audit ISP-reported broadband deployment by combining FCC broadband availability data with crowdsourced network measurements. [*Paper under submission to IMC '27*] [*Poster at TPRC54*]
- Built longitudinal data pipelines to evaluate claimed broadband infrastructure upgrades across technologies, providers, and geographic regions, and identified cases where reported deployment did not align with measured network performance.

2. Broadband Surveyor Tool (BBS): Surveying Long-Tail ISPs.

- Developing the Broadband Surveyor Tool (BBS) to systematically measure broadband service availability and affordability across a large and diverse set of providers. [*Paper under submission to IMC '27*]

3. Capturing Internet Service Provider (ISP) Footprints.

- Investigated multi-modal spatio-temporal representations of broadband infrastructure by combining broadband availability, geospatial, and network measurement datasets to characterize ISP footprints across geographic regions and over time. [*Poster*]

Project Scientist, Indian Institute of Technology (IIT), Delhi. *Jan '24 - June '25*
ICTD Lab
PI: Prof. Aaditeshwar Seth

1. A methodology to assess equity in access to water in villages in India.

- Developed a methodology for assessing equity in access to water resources and enhancing agroforestry for agricultural income improvement. [*Paper in preparation*] [Code]
- Trained object detection and image segmentation models on high resolution imagery to geotag agricultural assets (ponds, wells, crop fields, and plantation trees) from Google Maps. Applied entropy-based methods and CutMix Augmentation techniques to reduce false positive cases. [Paper] [Code]

2. Tracking Socioeconomic Development in Rural India Using Satellite Imagery. [Paper] [code]

- Aggregated census variables at the village level to create five socioeconomic indicators covering 75% of rural villages in India.
- Designed and trained a model that combines daytime satellite imagery and nightlight data to create a proxy for census, enabling the study of socioeconomic development trends and dynamics in rural villages across India.

3. Created administrative boundary shapefiles at multiple levels (village, block, subdistrict, state) for 15 Indian states. [Code]

Research Assistant, Ashoka University. *Sept '24 - Present*
PI: Prof. Anirban Sen and Prof. Debayan Gupta
1. Leveraging LLMs in Generating Stories Around Indian Policies. [Paper] [Code] [Website]

- Collected longitudinal data from mass media sources on various Indian policy issues (Annual Budget, Farmers' Protest, Unlawful Activities (Prevention) Act, and Big Corporations) between 2014 and 2024.
- Fine-tuned open-source LLMs to enable question-answering on policy issues and generate engaging, factually accurate stories around them. Additionally, evaluated the effectiveness of storytelling with LLMs through a human-in-the-loop approach.

Project Scientist, IIT Delhi. *Jun '22 - Nov '23*
Vision Lab
PI: Prof. Chetan Arora and Prof. Nandana Sengupta
Assessing the Feasibility and Ethics of Economic Status Prediction using Deep Learning on Household Images. [Paper] [Code]

- Investigated whether household images collected from remote tribal areas of four northeastern Indian states, could accurately predict their socioeconomic status thus allowing efficient allocation of resources at a granular level.
- Trained Transformer-Based vision models using metric learning based losses. Conducted two benchmarking analyses: with models trained only on the asset information of the households collected during the same survey and with human accuracy.

Master's Dissertation, South Asian University. *May '21 - May '22*
Machine Learning and Statistical Inference Lab
PI: Prof. Reshma Rastogi
Generative Adversarial Networks and its Applications. [Paper] [Code]

- Executed GAN on MNIST Digit dataset and MNIST Fashion dataset with Self Attention Module and Spectral Normalization in both Generator and Discriminator. Added Two Time Scale Update to stabilize the training of the GAN.
- Outcome of my dissertation was a novel architecture called MLGAN (Multi-Label Generative Adversarial Network), which was designed to address the challenge of data imbalance in multilabel datasets.

Industry Experience

Sentiment Analysis Internship, **Omdena** *Sept '21 - Oct '21*
Business and Data Analytics Internship, **The Sparks Foundation** *Aug '21 - Sep '21*
Big Data and ML Internship, **Rani.ai** *Aug '21 - Oct '21*

SERVICE

- Reviewer, ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies (COMPASS '25) *2025*